

Force and laws of motion: Source questions

IIT-JEE · 2007 · Paper I · Q3

Official past paper, faithful restatement of the original

Subject: Physics

Topic: Laws of Motion

Concept: Net force, not any one force, sets acceleration

Concept difficulty: Medium concept

Question difficulty: Multi-step reasoning

Family: Newton-law and observer-frame practice

Question type: Single-correct multiple choice

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Provenance: Verified past-exam adaptation

Two equal masses m lie on a smooth horizontal surface and are joined by a light string of total length $2a$. The midpoint of the initially taut string is pulled vertically upward by a small constant force F . When the masses are separated by $2x$, determine the magnitude of the acceleration of either mass.

Conditions

- Horizontal surface is frictionless.
- String is light and remains taut.
- Each half-string has length a .

(A) $(F/2m) \cdot a / \sqrt{a^2 - x^2}$

(B) $(F/2m) \cdot x / \sqrt{a^2 - x^2}$

(C) $(F/2m) \cdot x / a$

(D) $(F/2m) \cdot \sqrt{a^2 - x^2} / x$

Hints from the source

1. How does the vertical displacement relate to half-string length and half-separation?